



Solids, Liquids & Gases

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RINGKASAN MATERI & TARGET KOMPETENSI

Comprehensive study of the kinetic particle theory of matter, states of matter (solids, liquids, gases), phase changes and heating/cooling curves with latent heat plateaus, gas pressure and temperature effects, and rates of diffusion in liquids and gases relative to molecular mass (M_r).



1. The Kinetic Particle Theory & States of Matter

Extended tier only

The Kinetic Particle Theory of Matter

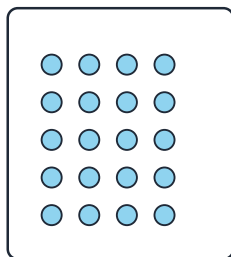
All matter is composed of extremely small particles (atoms, molecules, or ions) that are in **continuous, random motion**. The physical properties of a substance depend on:

1. The **arrangement** of its particles.
2. The **movement** of its particles.
3. The **forces of attraction** holding the particles together.

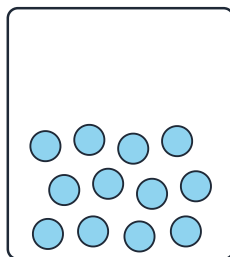
Comparative Summary of the Three States of Matter

Property	Solid	Liquid	Gas
Arrangement of particles	Regular lattice pattern	Random / irregular arrangement	Random / irregular arrangement
Movement of particles	Vibrate about fixed positions	Move and slide past one another	Move rapidly and randomly in all directions
Closeness / Spacing	Packed very closely together	Close together (touching)	Very far apart (large empty spaces)
Shape & Volume	Fixed shape and fixed volume	Fixed volume, adopts container shape	No fixed shape, expands to fill container
Relative Density	High density	Medium to high density (water is an exception)	Very low density
Compressibility	Virtually incompressible	Virtually incompressible	Easily compressed (lots of empty space)
Kinetic Energy	Lowest kinetic energy	Greater kinetic energy	Highest kinetic energy (approx. 500 m/s at room temperature)

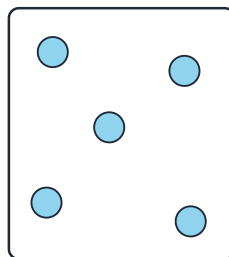
Particle Models in Solids, Liquids, and Gases



SOLID (REGULAR)



LIQUID (RANDOM)



GAS (FAR APART)



Examiner Tips and Tricks: Cambridge Examiner Tips & Tricks: Key Keywords

When answering exam questions about particle theory, Cambridge examiners award marks strictly based on precise terminology:

Solid: Particles are in a **fixed regular lattice** and only **vibrate about fixed positions**.

Liquid: Particles are **randomly arranged**, still **touching/close**, and **move/slide over each other**.

Gas: Particles are **randomly arranged**, **far apart**, and move **rapidly in all directions**.

 Avoid vague words like *"move around freely"* for liquids — liquids are confined to the bottom of containers!

2. State Changes & Phase Transitions

Extended tier only

Interconversion Between States of Matter

State changes are **physical changes**: no new chemical bonds are broken or formed, and the chemical composition remains identical.



- **Melting:** Solid $\rightarrow$ Liquid. Thermal energy increases particle kinetic energy until vibrations overcome the lattice forces. Occurs at the **melting point (m.p.)**.
- **Freezing:** Liquid $\rightarrow$ Solid. Reverse of melting. Occurs at the **exact same temperature** as melting (pure water melts and freezes at 0°C).
- **Boiling vs Evaporation:**
- **Boiling:** Takes place **throughout the liquid** at a single, fixed temperature called the **boiling point (b.p.)**. Gas bubbles form below the surface.
- **Evaporation:** Takes place **only at the surface** of the liquid over a **wide range of temperatures** below the boiling point. Fast-moving particles at the surface escape into the vapor phase.
- **Condensation:** Gas $\rightarrow$ Liquid upon cooling. Particles lose kinetic energy and cluster together.
- **Sublimation:** Solid $\rightarrow$ Gas directly without passing through the liquid state (e.g., dry ice $\text{CO}_2(\text{s})$, iodine $\text{I}_2(\text{s})$).
- **Desublimation (Deposition):** Gas $\rightarrow$ Solid directly (e.g., formation of frost).

Key Examiner Notes

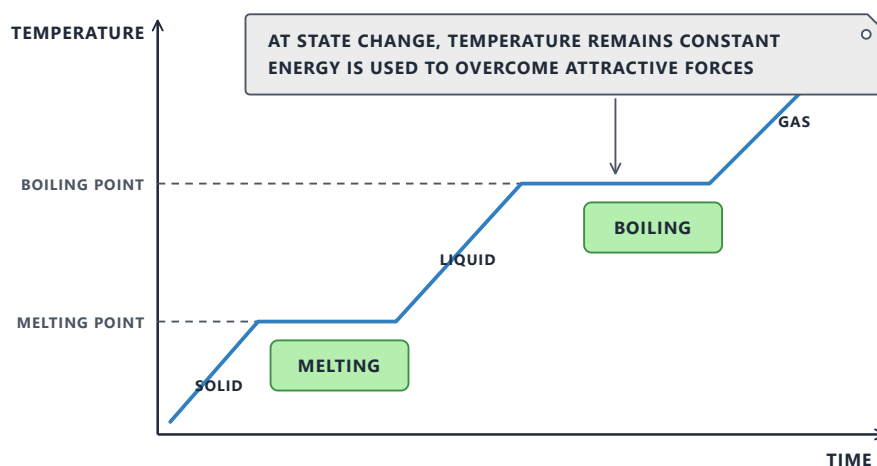
Cambridge exam questions often display phase changes with the reversible sign $\rightleftharpoons$. This indicates that state transitions are completely physical and can be reversed by adding or removing thermal energy.

3. State Changes & Kinetic Theory (Extended Tier)

Extended tier only

Heating Curves & Temperature Plateaus (Extended Tier)

When a pure substance is heated at a steady rate, its temperature changes can be plotted against time.



? Why Does Temperature Remain Constant During Melting & Boiling?

This is a classic **2-mark question** in Cambridge Paper 4:

- Mark 1:** Temperature is a measure of the average kinetic energy of the particles.
- Mark 2:** During a state change, the thermal energy absorbed is **not** used to increase kinetic energy. Instead, it is used to **overcome the attractive intermolecular forces** between particles.
- Therefore, because kinetic energy does not increase, the **temperature stays constant** until the entire sample has completed the phase transition.



Key Examiner Notes & Exam Priorities: ❄️ Cooling Curves: The Mirror Image

On a cooling curve, the plateaus correspond to **condensation** and **freezing**. Thermal energy is released as attractive bonds form between particles, keeping the temperature constant until the entire liquid has frozen.

4. Pressure & Temperature in Gases

Extended tier only

👉 How Is Gas Pressure Created?

Gaseous particles are in continuous, rapid, random motion.

- Gas pressure is produced when **gas particles collide with the interior walls of their container**.
- Each collision exerts a small force against the wall. The total force divided by the area of the wall equals the **pressure** ($P = \frac{F}{A}$).

🌡️ Effect of Temperature on Gas Volume & Pressure

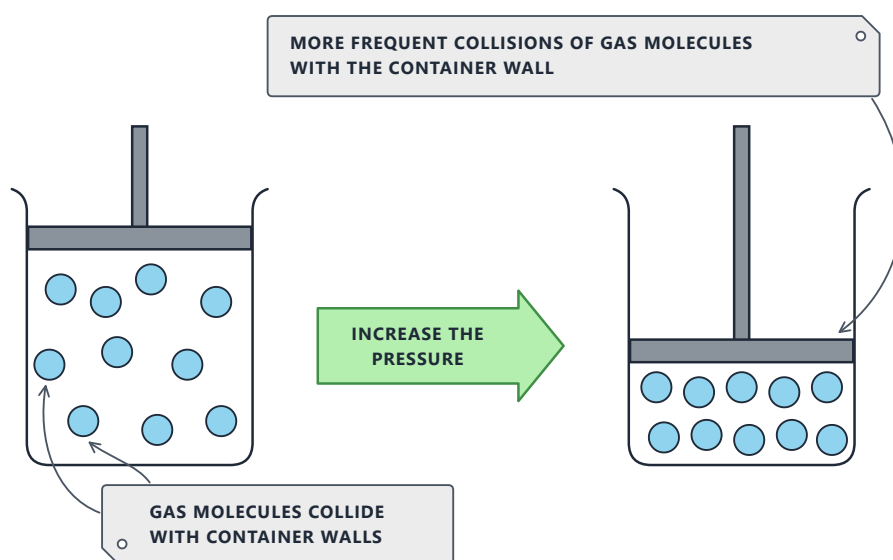
1. Increasing Temperature:

- Thermal energy transforms into **kinetic energy**. Particles move faster.
- Particles collide with container walls **more frequently** and with **greater force**.
- If the container is flexible (e.g. a hot air balloon or piston), the walls expand, meaning **volume increases**.
- If the container is rigid (fixed volume), **pressure increases**.

🔧 Effect of Pressure on Gas Volume (Compression)

When a gas is squeezed inside a cylinder using a piston:

- Decreasing volume packs the **same number of particles** into a **smaller space**.
- Particles collide with container walls **more frequently**, leading to **higher pressure**.





Examiner Warning & Common Misconceptions: Cambridge Examiner Trap: "Particles DO NOT Expand!"

A major reason students lose marks in Cambridge exams:

Wrong statement: *"The gas particles expand when heated."* (0 marks)

Correct statement: *"The gas particles gain kinetic energy, move faster, and spread further apart. As a result, the **volume of the gas** (the material) expands."*
(Full marks)



5. Diffusion & Molecular Mass (Extended Tier)

Extended tier only

What Is Diffusion?

Diffusion is the net movement of particles from an area of **high concentration** to an area of **low concentration** down a concentration gradient, as a result of their random motion.

- Occurs in **gases and liquids** (fluids where particles can move freely).
- Requires **no energy input** (passive physical process).
- Occurs **faster at higher temperatures** because particles have higher kinetic energy.

Evidence for Diffusion in Liquids: KMnO_4 Crystals

When a small crystal of deep-purple potassium manganate(VII) (KMnO_4) is placed at the bottom of a beaker of water:

1. Initially, water around the crystal turns intensely purple (high concentration).
2. Random collisions between moving water molecules and manganate ions gradually disperse the color.
3. After several hours, the entire solution becomes an **evenly colored pale purple solution** (equilibrium reached).

Evidence for Diffusion in Gases: Bromine Gas

A gas jar of orange-brown bromine gas is inverted under a jar of air separated by a glass cover:

- When the cover is removed, the brown gas spreads upwards into the air jar, while colorless air diffuses downward.
- After about 5 minutes, both jars are an even orange-brown color throughout.
- Diffusion is **much faster in gases than in liquids** because gas particles have far greater speeds (approx. 500 m/s) and plenty of empty space to move through.

How Molecular Mass (M_r) Affects Rate of Diffusion (Extended Tier)

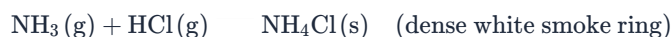
At the same temperature, different gas particles have the same average kinetic energy:

$$E_k = \frac{1}{2} mv^2$$

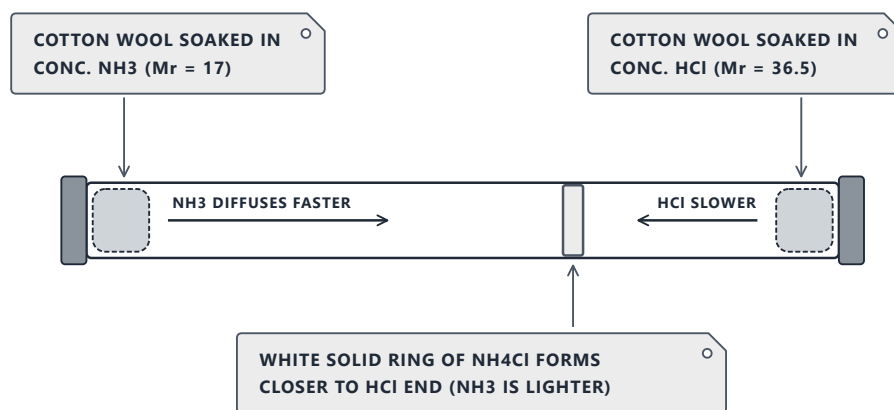
Consequently:

- Gases with a **lower relative molecular mass (M_r)** are lighter and **travel faster** (higher diffusion rate).
- Gases with a **higher relative molecular mass (M_r)** are heavier and **travel slower** (lower diffusion rate).

The Classic Experiment: Diffusion of NH_3 and HCl



- Relative molecular mass of Ammonia (NH_3): $M_r = 14 + (3 \times 1) = 17$
- Relative molecular mass of Hydrogen Chloride (HCl): $M_r = 1 + 35.5 = 36.5$



Key Observations:

1. The white ring forms **nearer to the hydrochloric acid end**.
2. **Reason:** Ammonia particles are lighter ($M_r = 17$) than hydrogen chloride particles ($M_r = 36.5$). Therefore, NH_3 particles **diffuse faster** and travel a **further distance** in the same time compared to HCl .

6. Cambridge Exam-Style Questions & Mark Schemes

Extended tier only · Analisis Langkah Solutif

Contoh Soal & Pembahasan Terpandu

[OSN / IChO Standards]

Question 1 (Paper 2 Multiple Choice — 1 Mark)

A student placed a porous pot containing hydrogen gas (H_2 , $M_r = 2$) into a beaker of air ($M_r \approx 29$). What happens to the water level in the manometer connected to the porous pot?



Examiner Tips and Tricks: Step-by-Step Mark Scheme Breakdown

1. Compare molecular masses: H_2 ($M_r = 2$) is far lighter than air ($M_r \approx 29$).
2. H_2 molecules diffuse **out** of the porous pot into the beaker much faster than air molecules diffuse **into** the pot.
3. As gas leaves the pot faster than it enters, pressure inside the pot **decreases**.
4. The liquid in the manometer limb closest to the porous pot **rises**.

Question 2 (Paper 4 Structured Theory — 4 Marks)

A sealed glass tube is set up with cotton wool soaked in concentrated aqueous ammonia at one end and concentrated hydrochloric acid at the other end.

(a) State the name of the white solid that forms in the tube. [1]

> **Mark Scheme:** Ammonium chloride (allow NH_4Cl).

(b) Explain why the white solid forms closer to the hydrochloric acid end of the tube. [3]

> **Mark Scheme:**

> - **Mark 1:** Ammonia has a lower relative molecular mass ($M_r = 17$) than hydrogen chloride ($M_r = 36.5$).

> - **Mark 2:** Ammonia molecules diffuse / move **faster** than hydrogen chloride molecules.

> - **Mark 3:** Ammonia molecules travel a **greater distance** in the same time.

Checkpoint Pemahaman & Miskonsepsi

Extended tier only · Latihan Mandiri

SOAL #1 (1. The Kinetic Particle Theory & States of Matter)

Which row correctly describes the movement and arrangement of particles in a liquid?

A.

Movement: vibrate about a fixed position | Arrangement: regular lattice

B.

Movement: slide past one another | Arrangement: random / irregular

C.

Movement: move rapidly in all directions | Arrangement: regular lattice

D.

Movement: stationary | Arrangement: random / irregular

Kunci & Penjelasan Konseptual:

In liquids, particles are close together in an irregular (random) arrangement, but they have enough energy to slide and roll over one another, allowing liquids to flow.

SOAL #2 (1. The Kinetic Particle Theory & States of Matter)

Gases have a much lower density than liquids because gas particles themselves are smaller in size than liquid particles.

Kunci & Penjelasan Konseptual:

False! The particles themselves do NOT change in size. Gases have a low density because the particles are separated by vast amounts of empty space, meaning far fewer particles occupy a given volume.

SOAL #3 (2. State Changes & Phase Transitions)

Which statement describes the main difference between boiling and evaporation?

A.

Boiling occurs at any temperature, whereas evaporation occurs only at the boiling point

B.

Boiling occurs throughout the liquid at a fixed temperature, whereas evaporation occurs only at the liquid surface over a range of temperatures

C.

Evaporation produces gas bubbles beneath the surface, whereas boiling does not

D.

Evaporation is a chemical change, whereas boiling is a physical change

Kunci & Penjelasan Konseptual:

Boiling happens throughout the entire bulk of the liquid at a precise boiling point, producing vapor bubbles. Evaporation occurs exclusively at the liquid surface over a continuous range of temperatures below the boiling point.

SOAL #4 (3. State Changes & Kinetic Theory (Extended Tier))

A sample of pure solid substance X is heated until it becomes a gas. Why does the temperature graph show a flat horizontal line while melting?

A.

The heating supply was temporarily turned off

B.

Thermal energy is being used to break/overcome attractive forces between particles, rather than increasing their kinetic energy

C.

The particles stop vibrating during melting

D.

The substance has reached absolute zero

Kunci & Penjelasan Konseptual:

During melting, all added thermal energy goes into overcoming the intermolecular forces holding the lattice together. Since temperature measures average kinetic energy, temperature remains completely flat until all solid has melted.

SOAL #5 (4. Pressure & Temperature in Gases)

When gas inside a syringe is heated, the individual gas particles increase in size, causing the syringe plunger to push outwards.

Kunci & Penjelasan Konseptual:

False! Particles NEVER expand or shrink. Heating increases particle speed and kinetic energy, so they collide more vigorously with the plunger and spread further apart, increasing the macroscopic volume.

SOAL #6 (5. Diffusion & Molecular Mass (Extended Tier))

Four gas samples are kept at the same temperature: Helium (He, $M_r=4$), Methane (CH_4 , $M_r=16$), Oxygen (O_2 , $M_r=32$), and Sulfur Dioxide (SO_2 , $M_r=64$). Which gas diffuses fastest?

A.

Helium (He)

B.

Methane (CH_4)

C.

Oxygen (O_2)

D.

Sulfur Dioxide (SO_2)

Kunci & Penjelasan Konseptual:

According to kinetic theory, rate of diffusion is inversely proportional to the square root of relative molecular mass. Helium has the lowest M_r (4), so its particles travel with the highest average speed.